YEAR 10 PHYSICS, AQA GCSE, LESSON 8 OF 8

PARTICLE MODEL OF MATTER

LESSON 08

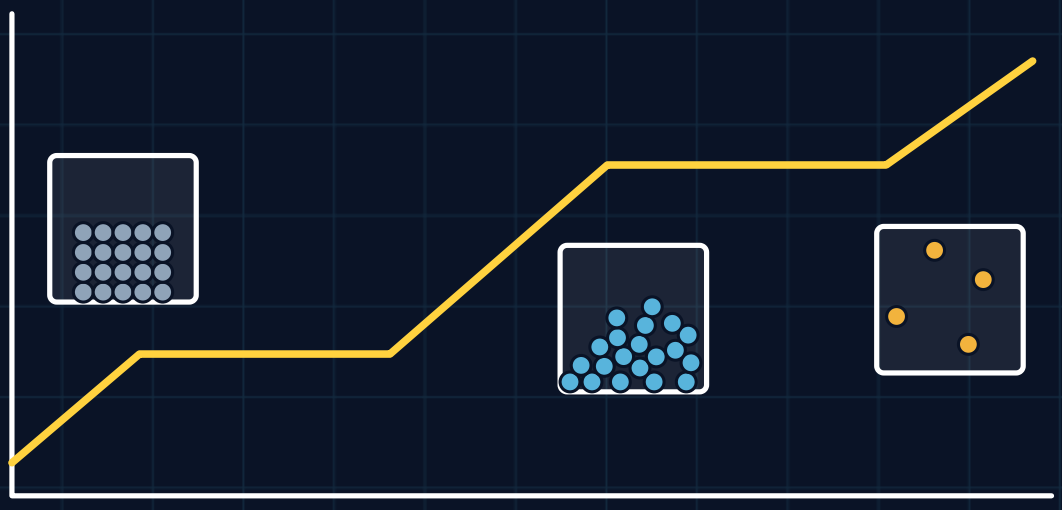
MEASURING SPECIFIC HEAT CAPACITY (REQUIRED PRACTICAL)

How could you find out which metal a block is made of, using only a heater and a thermometer?

IN THIS LESSON YOU WILL

- describe how to measure the specific heat capacity of a metal block
- calculate the energy supplied with $E = VIt$, or read it from a joulemeter
- calculate the specific heat capacity from the results
- explain why the result is usually too high, and how to improve it

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 8** of the Particle Model of Matter unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW

Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT

The explanation and diagrams. Fill in each example from the slides with your teacher, then try “Your turn” alone. Read this again when you revise.

TRY IT

Practice that builds up: first with support, then on your own, then a challenge. Every second lesson ends with the Quick check from the slides.

PRACTICAL

Record your measurements and work out results from them.

EXAM CORNER

AQA-style questions with a tip on how the marks are given.

REVISE IT

Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are close together **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

01 Solids, liquids and gases

02 Gas particles and pressure

03 Density

04 Measuring density (required practical)

05 Internal energy and heating curves

06 Specific latent heat

07 Specific heat capacity

08 **Measuring specific heat capacity (required practical)**
Heating a metal block, $E = V I t$ or a joulemeter, the graph method, why the answer comes out too high (slides 40–44)

The **equation toolkit** near the back has every equation in the unit, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-model**

Lesson

08

MEASURING SPECIFIC HEAT CAPACITY (REQUIRED PRACTICAL)

AQA Physics 4.1.1.3, required practical 1; Combined Science 6.1.1.3, required practical 14

BIG QUESTION **How could you find out which metal a block is made of, using only a heater and a thermometer?**

BY THE END YOU CAN

- describe how to measure the specific heat capacity of a metal block
- calculate the energy supplied with $E = V I t$, or read it from a joulemeter
- calculate the specific heat capacity from the results
- explain why the result is usually too high, and how to improve it

immersion heater **joulemeter** **insulation** **energy supplied** **gradient** **systematic error**

YOU WILL NEED

Per group: 1 kg metal block with two holes, 12 V immersion heater, thermometer, insulation jacket and heatproof mat, power supply, ammeter, voltmeter, leads, stopwatch, balance. A joulemeter if you have one. A drop of oil for the thermometer hole.

Teaching deck, slides 40–44
panphy.app/gcsephy/decks/particle-model/#40

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Define specific heat capacity.

The energy needed to raise the temperature of 1 kg of a substance by 1 °C.

2
Write the two equations for electrical power and energy from Electric Circuits.

$P = V I$ and $E = P t$

3
Convert 10 minutes to seconds.

10 × 60 = 600 s

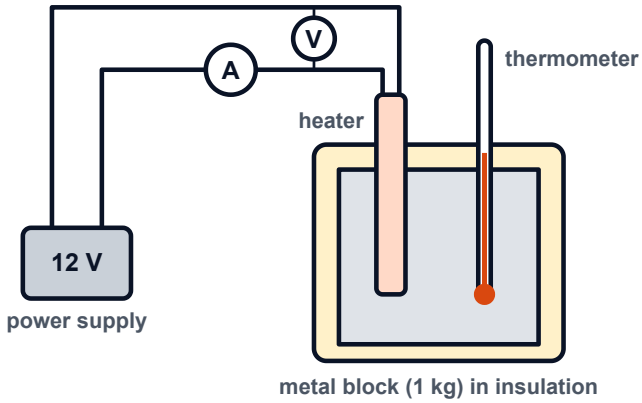
EXPLAIN

ENERGY IN, TEMPERATURE UP

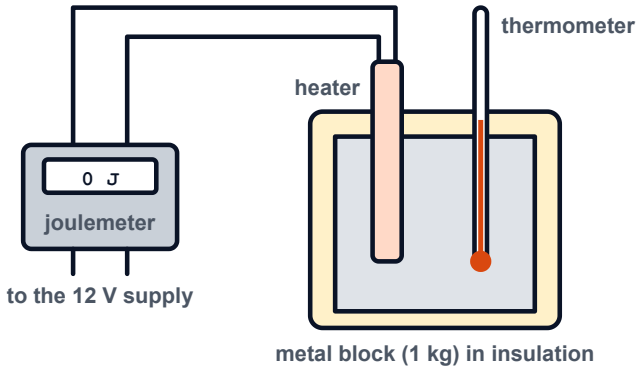
To find a specific heat capacity you need three measurements: the **mass** of the block, its **temperature rise** and the **energy supplied** to it. Then $c = \dfrac{\Delta E}{m\,\Delta\theta}$.

An electric heater is used, not a Bunsen burner, because the energy that an electric heater supplies can be measured.

TWO WAYS TO MEASURE THE ENERGY SUPPLIED



Ammeter in series, voltmeter across the heater



A joulemeter reads the energy directly

$$E = V\,I\,t$$

energy supplied = potential difference × current × time

E in J, V in V, I in A, t in s

This joins two equations you know: $P = V\,I$ and $E = P\,t$.

WORKED EXAMPLE

A heater runs at 12 V and 4.0 A for 5.0 minutes. Calculate the energy supplied.

CONVERT	$t = 5.0 \times 60 = 300$ s
EQUATION	$E = V\,I\,t$
SUBSTITUTE	$E = 12 \times 4.0 \times 300$
ANSWER	$E = 14\,400$ J

YOUR TURN

A heater runs at 12 V and 3.0 A for 10 minutes. Calculate the energy supplied.

CONVERT	$t = 10 \times 60 = 600$ s
EQUATION	$E = V\,I\,t$
SUBSTITUTE	$E = 12 \times 3.0 \times 600$
ANSWER	$E = 21\,600$ J

WHY THE ANSWER COMES OUT TOO HIGH

Not all of the energy supplied stays in the block. Some is transferred to the surroundings, and some heats the heater itself. So the temperature rise is **smaller** than it should be, and $c = \dfrac{\Delta E}{m\,\Delta\theta}$ comes out **too big**.

- **Insulate** the block (sides, top and underneath) to reduce energy transfer to the surroundings.

- Put a drop of **oil** in the thermometer hole, so the thermometer is in good thermal contact with the block.
- After switching off, keep watching: record the **highest** temperature the thermometer reaches.

INVESTIGATE

REQUIRED PRACTICAL: SPECIFIC HEAT CAPACITY

The heater and block get hot. Do not touch them until they have cooled.

STAY SAFE

- The immersion heater gets very hot: never switch it on outside the block.
- Stand the block on a heatproof mat and leave it to cool before you move it.
- Do not exceed 12 V. Switch off at the power supply before you change the circuit.

METHOD

1. Measure the mass of the block with a balance.
2. Wrap the block in insulation. Put the heater in the large hole and the thermometer, with a drop of oil, in the small hole.
3. Connect the ammeter in series with the heater and the voltmeter across it (or connect a joulemeter).
4. Record the starting temperature. Switch on and start the stopwatch.
5. Record the potential difference and the current. Record the temperature every 2 minutes for 10 minutes.
6. Switch off. Record the highest temperature reached.

1. Record your measurements.

Material of block	Mass / kg	Potential difference / V	Current / A
aluminium	1.00	12.0	4.0

2. Complete the table. Energy supplied = $V\,I\,t$, or read the joulemeter.

Time / s	0	120	240	360	480	600
Energy supplied / J	0	5760	11520	17280	23040	28800
Temperature / °C	20.0	24.6	30.8	37.0	43.0	49.0

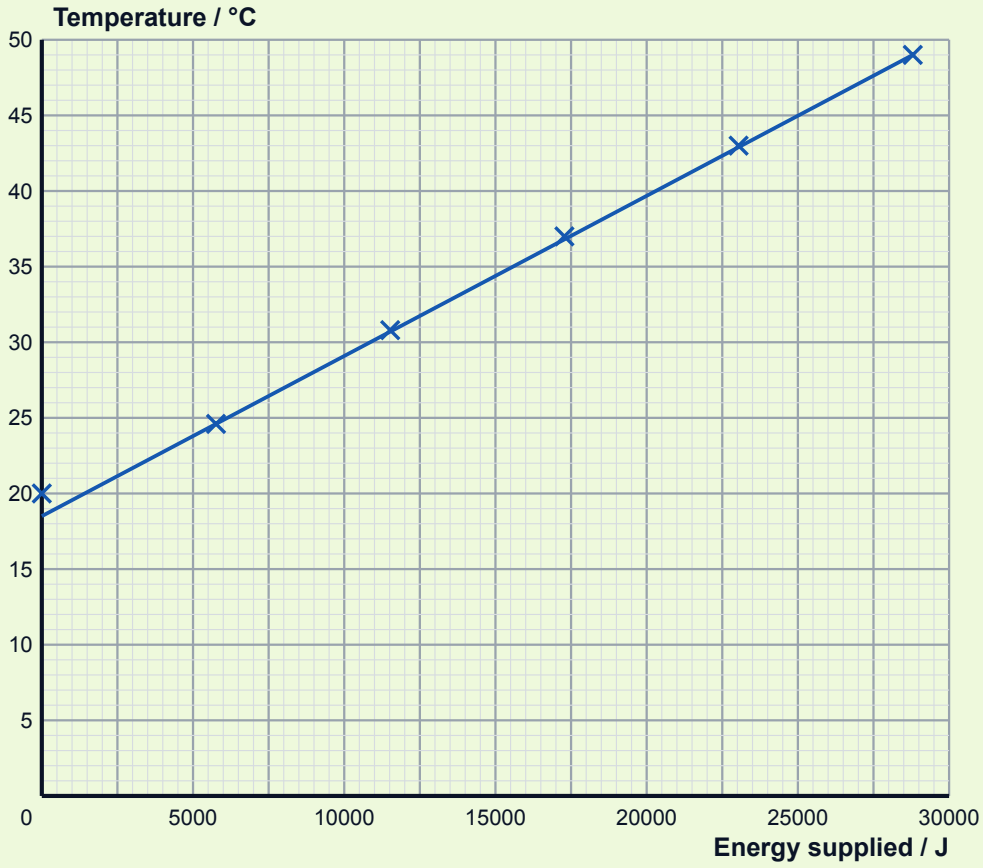
RESULTS: FROM THE TOTALS

3 Calculate the specific heat capacity of the block from the total energy supplied and the total temperature rise.

TEMPERATURE RISE	$\Delta\theta = 49.0 - 20.0 = 29.0$ °C
EQUATION	$\Delta E = m\,c\,\Delta\theta$
REARRANGE	$c = \dfrac{\Delta E}{m\,\Delta\theta}$
SUBSTITUTE	$c = \dfrac{28\,800}{1.00 \times 29.0}$
ANSWER	**993 J/kg °C (about 990)** unit:

RESULTS: FROM A GRAPH

4 Plot temperature against energy supplied. Draw a straight line of best fit through the part of the graph that is straight.



5 Find the gradient of your line. Then use $c = \dfrac{1}{m \times \text{gradient}}$

Gradient $= \dfrac{49.0 - 24.6}{28\,800 - 5760} = 0.00106$ **°C/J.** $c = \dfrac{1}{1.00 \times 0.00106} = 940$ **J/kg °C**

6 The graph curves at the start. Suggest why.

At first the heater itself is warming up, and it takes time for energy to spread through the block to the thermometer.

PRACTISE

TRY IT

7 The accepted specific heat capacity of aluminium is 900 J/kg °C. Is your value higher or lower? Explain why.

Higher. Some of the energy supplied was transferred to the surroundings instead of the block, so the temperature rise was smaller than it should have been. Dividing by a smaller temperature rise gives a bigger value.

8 Draw a ring around the correct answer in each box.

The ammeter is connected | **in series with the heater** | **in parallel with the heater**

Insulation makes the measured specific heat capacity | **further from the true value** | **closer to the true value**

Energy transferred to the surroundings is | **a random error** | **a systematic error**

9 Explain why the gradient method gives a better value than using only the first and last readings.

The line of best fit uses all the readings, and it leaves out the curved start, where the heater and block were still warming up unevenly.

10 **CHALLENGE** A student repeats the practical with 1.0 kg of water in an insulated beaker, with the same heater for the same time. Predict how the temperature rise compares with that of the aluminium block, and explain.

The temperature rise of the water is much smaller (about a fifth), because water has a much higher specific heat capacity (4200 compared with 900 J/kg °C): more energy is needed for each degree.

CHECKPOINT

QUICK CHECK

Slide 44, Lessons 7–8. Answer each question, then check it as your teacher reveals the answer.

1
State the unit of specific heat capacity.

J/kg °C

2
Water is heated from 15 °C to 95 °C. Calculate $\Delta\theta$.

80 °C

3
Calculate the energy to heat 3.0 kg of iron by 50 °C. c = 450 J/kg °C

$3.0 \times 450 \times 50 = 67\,500$ **J**

4
A heater runs at 12 V and 3.0 A for 300 s. Calculate the energy supplied.

$E = V\,I\,t = 10\,800$ **J**

5
Why is the metal block insulated?

So that less energy is transferred to the surroundings.

6
Ice melting at 0 °C: do you use c or L?

L: the state changes, not the temperature.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Name what is measured and with what: mass (balance), temperature (thermometer), energy (joulemeter, or ammeter, voltmeter and stopwatch).

11 Describe a method to determine the specific heat capacity of a metal block. You should include the measurements needed and how they are used. **[6 marks]**

Indicative content: measure the mass of the block with a balance; insulate the block; put an immersion heater and a thermometer in the block; record the starting temperature; switch on the heater for a measured time; measure the energy supplied with a joulemeter (or measure p.d., current and time and use E = V I t); record the highest temperature and calculate the temperature rise; use c = ΔE ÷ (m Δθ); repeat, or plot temperature against energy and use the gradient. Level 3 (5–6): a detailed, logically ordered method that would give a valid result.

12 A 2.0 kg copper block is heated with an immersion heater. The joulemeter reads 16 400 J and the temperature of the block rises from 21 °C to 41 °C. Calculate the specific heat capacity of copper given by these results. **[4 marks]**

TEMPERATURE RISE	$\Delta\theta = 41 - 21 = 20$ °C (1)
EQUATION	$\Delta E = m\,c\,\Delta\theta$
REARRANGE	$c = \dfrac{\Delta E}{m\,\Delta\theta}$ (1)
SUBSTITUTE	$c = \dfrac{16\,400}{2.0 \times 20}$ (1)
ANSWER	**410 J/kg °C (1)** unit:

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Measure the mass (balance), the temperature rise (thermometer) and the energy supplied.
- Energy supplied: a joulemeter, or $E = V I t$ with an ammeter, a voltmeter and a stopwatch.
- $c = \dfrac{\Delta E}{m \Delta \theta}$, or from the gradient of a temperature–energy graph.
- Energy transferred to the surroundings makes the result too high. Insulation reduces this.

COVER THE PAGE AND ANSWER

1 Which three measurements are needed to find a specific heat capacity?

2 Why is the block wrapped in insulation?

3 Why is the measured value usually too high?

4 Write the equation for the energy supplied by a heater, using V, I and t.

I CAN…

- describe the method for measuring specific heat capacity
- calculate the energy supplied with $E = V I t$
- calculate specific heat capacity from results
- use the gradient of a temperature–energy graph
- explain why the result is too high and how to improve it

REFERENCE

EQUATION TOOLKIT

Every equation in the unit. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find V from $\rho = \frac{m}{V}$, cover V. m is above ρ, so $V = \frac{m}{\rho}$.

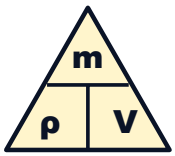
$\rho = \frac{m}{V}$

Lesson 3

density = mass ÷ volume

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$



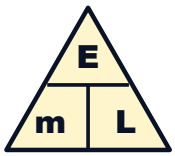
$E = m L$

Lesson 6

energy for a change of state = mass × specific latent heat

J, kg, J/kg

$L = \frac{E}{m}$ and $m = \frac{E}{L}$



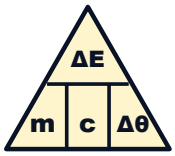
$\Delta E = m c \Delta\theta$

Lesson 7

change in thermal energy = mass × specific heat capacity × temperature change

J, kg, J/kg °C, °C

$c = \frac{\Delta E}{m \Delta\theta}$ and $\Delta\theta = \frac{\Delta E}{m c}$



$E = V I t$

Lesson 8

energy supplied = potential difference × current × time

J, V, A, s

From $P = V I$ and $E = P t$ (Electric Circuits)

1
Write what you know. Convert first: g → kg, cm³ → m³, kJ → J, min → s.

2
Write the equation in symbols.

3
Rearrange if you need to, then substitute the numbers.

4
Give the answer with a unit. Ask: is it sensible?

REFERENCE

KEY WORDS FROM THIS LESSON

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

Joulemeter: A meter that measures the energy supplied to an electrical appliance

Insulation: Material that reduces energy transfer to the surroundings

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. The energy needed to raise the temperature of 1 kg of a substance by 1 °C.
2. $P = V I$ and $E = P t$
3. 10 × 60 = 600 s

COVER THE PAGE

1. Mass, temperature rise and energy supplied.
2. To reduce energy transfer to the surroundings.
3. Some of the energy supplied goes to the surroundings, so the temperature rise is too small.
4. $E = V I t$